RRC-Raptor-V4H

- GPIO

V0.0.1

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i. VERSION LIST

Description	Version	Date	Author
GPIO	V0.0.1	2024/06/28	HowardChao

ii. HASHTAG

#V4H
#BSP
#ubuntu
#GPIO

1. Overview

Target：Introduce how to use GPIO function on V4H-Raptor.

2. GPIO

The position of GPIO connector (CN29) and its pinout are shown in Figure 1 and Figure 2. The pin functions are listed below in Table 1.

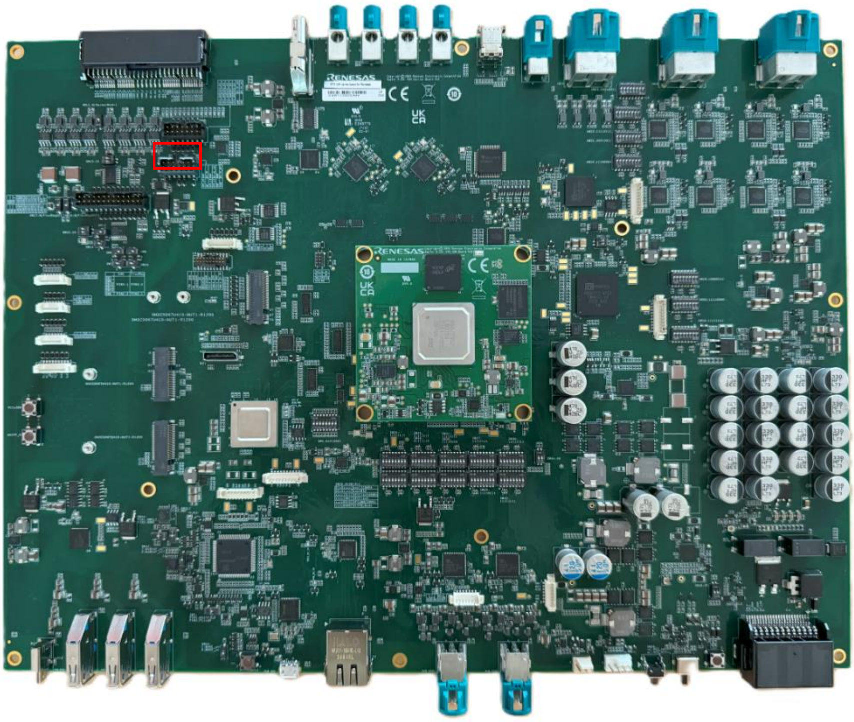

Figure 1. The position of GPIO connector

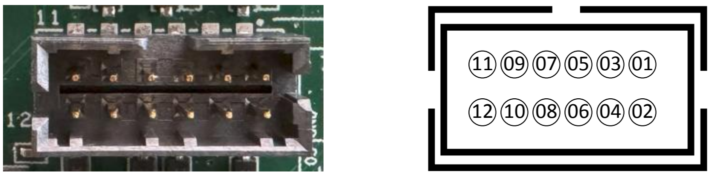

Figure 2. Pinout of GPIO connector

Table 1. Pin Functions of GPIO connector

Function	PIN	PIN	Function
D3.3V_SPI	1	2	IRQ2/MD0_SW (GP0_04)
GND	3	4	GND
D3.3V_I2C	5	6	GP8_12_V
GND	7	8	GP8_13/MD27_SW
D3.3V	9	10	NC
GND	11	12	GP6_05/AVB1_3V3

GPIO commands

There are several commands to use for GPIO. The following describes the functions of each command.

```
$ gpio
gpiodetect    gpiofind      gpioget       gpioinfo      gpiomon       gpioset
```

2.1. gpiodetect

This command lists all available GPIO chips in the system.

```
$ gpiodetect
gpiochip0 [e6050180.gpio] (19 lines)
gpiochip1 [e6050980.gpio] (29 lines)
gpiochip10 [12-0073] (8 lines)
gpiochip11 [13-0070] (8 lines)
gpiochip12 [13-0073] (8 lines)
gpiochip13 [14-0073] (8 lines)
gpiochip14 [14-0070] (8 lines)
gpiochip2 [e6058180.gpio] (20 lines)
gpiochip3 [e6058980.gpio] (30 lines)
…
gpiochip8 [e6068180.gpio] (14 lines)
gpiochip9 [12-0070] (8 lines)
```

2.2. gpiofind

This command is used to find the specific GPIO line associated with a particular function name. If the GPIO line has not been set or named, it will display “unnamed” by “gpioinfo” command.

```
$ gpiofind “USR1”
gpiochip0 24
```

2.3. gpioget

This command allows users to read the current level (high/low) of one or more GPIO lines to understand their status.

```
$ gpioget gpiochip0 4
0
$ gpioget gpiochip8 12 13
1 0
```

2.4. gpioinfo

This command can be used to see which GPIOs are in use, whether they are configured as input or output, and their active state (high/low). When followed by a number corresponding to a chip, the command will only list the GPIOs in that specific chip. If no parameters are provided, it will list all GPIOs in the system.

```
$ gpioinfo 8
gpiochip8 - 14 lines:
        line   0:    unnamed    unused   input   active-high
        …
        line  13:    unnamed    unused   input   active-high
$ gpioinfo
gpiochip0 - 19 lines:
        line   0:    unnamed    unused   input   active-high
        …
        line  18:    unnamed    unused   input   active-high
gpiochip1 - 29 lines:
        line   0:    unnamed    unused   input   active-high
        …
        line  28:    unnamed    unused   input   active-high
gpiochip10 - 8 lines:
        line   0:    unnamed    unused   output  active-high
        …
        line   7:    unnamed    unused   input   active-high
…
```

```
gpiochip14 - 8 lines:
        line   0:      unnamed       unused   input   active-high
        …
        line   7:      unnamed       unused   input   active-high
gpiochip2 - 20 lines:
        line   0:      unnamed       unused   input   active-high
        …
        line  19:      unnamed       unused   input   active-high
…
gpiochip9 - 8 lines:
        line   0:      unnamed       unused  output   active-high
        …
        line   7:      unnamed       unused   input   active-high
```

2.5. gpiomon

This command watches specified GPIO lines and outputs events when their level changes, allowing real-time monitoring of GPIO state changes.

```
$ gpiomon gpiochip8 12
event: FALLING EDGE offset: 14 timestamp: [1719814124.827479202]
event:  RISING EDGE offset: 14 timestamp: [1719814124.988156217]
event: FALLING EDGE offset: 14 timestamp: [1719814127.668793560]
event:  RISING EDGE offset: 14 timestamp: [1719814127.834825116]
```

2.6. gpioset

This command allows users to set the level (high/low) of one or more GPIO lines to control their state.

```
$ gpioset gpiochip8 12=1 (defaults to set value and exit immediately)
$ gpioset gpiochip8 12=1 13=1
$ gpioset –m time –s 10 gpiochip8 12=1 (set value and last for 10 seconds)
$ gpioset -m signal -b gpiochip8 12=1 (set value and wait for SIGINT/SIGTERM in
background )
```

3. REFERENCE